

Substitution, Expansion, Factorization

1 Substitution

- rewriting expressions
- simultaneous, sequential, syntactical substitution

2 Expansion and Factorization

- collecting terms in an expression
- numeric, exact, and symbolic factorization

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Introduction to Symbolic Computation
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Rewriting Expressions

Rewrite

$$(x + y)^2 + \frac{1}{(x + y)^2} \quad \text{into} \quad \frac{(x + y)^4 + 1}{(x + y)^2}.$$

Problem: bringing on the same denominator expands the numerator.

Solution:

- 1 replace $x + y$ by z ,
- 2 bring on the same denominator,
- 3 replace z by $x + y$.

The *substitution* of $x + y$ by z prevents the expansion.

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Simultaneous, Sequential, Syntactical Substitution

- ❶ Replace all variables at one: *simultaneous substitution*.

$$\begin{array}{ccc} a + 2b + 3c & & \\ \downarrow a=b & \downarrow b=c & \downarrow c=a \\ b + 2c + 3a & & \end{array}$$

- ❷ Replace variables one after the other: *sequential substitution*.

$$a + 2b + 3c \xrightarrow{a=b} 3b + 3c \xrightarrow{b=c} 6c \xrightarrow{c=a} 6a$$

- ❸ Replace in the string representation: *syntactical substitution*.

By default, substitutions respect the mathematical structure.

With strings, we can take slices and replace any character.

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Collecting Terms in an Expression

The product

$$(a + b + c) \star (x^2 + x + 1)$$

is not expanded automatically because of *expression swell*.

But we may want to collect terms as follows:

$$(a + b + c) x^2 + (a + b + c) x + (a + b + c).$$

Solved by two substitutions:

- 1 Substitute $a + b + c$ by z before expanding and then
- 2 substitute z by $a + b + c$ in the expanded expression.

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Numeric, Exact, and Symbolic Factorization

Factorization is the algebraic view on root finding.

- 1 Numeric is over the complex numbers, is approximate.
- 2 Exact is over the integers, rationals, or finite field.
- 3 Symbolic is with the addition of formal roots, algebraic numbers.

Be mindful of the *type casting* of the coefficients of the polynomial.

- 1 Numerically, the coefficients are cast into the complex numbers.
- 2 In exact factorization, we work with the same type of coefficients.
- 3 Symbolically, we cast the polynomial into the extended numbers.